

What This Experiment Actually Does

Instead of creating AWS resources, this Terraform code creates a simple local file:

sample.txt

with content:

Hello terraform

Purpose:

- Learn Terraform commands
- Understand IaC workflow
- No AWS account required

Smart choice for lab practical.

FULL EXPLANATION OF YOUR NOTES

Step 1 — Install Terraform

Download:

[Terraform Download Page](#)

Extract ZIP.

Move:

terraform.exe

to:

C:\Terraform

Step 2 — Add Terraform Path

Press:

Windows + R

Type:

sysdm.cpl

Press Enter.

Then:

Advanced

→ Environment Variables

→ System Variables

→ Path

→ Edit

→ New

Add:

C:\Terraform

Click OK.

This allows CMD to recognize Terraform commands globally.

Step 3 — Verify Terraform

Open CMD.

Run:

terraform-version

If version appears:

Terraform installed successfully.

Example:

Terraform v1.x.x

Step 4 — Create Project Folder

In CMD:

mkdir terraform-demo

cd terraform-demo

Now Terraform project folder is created.

Step 5 — Create Terraform File

Run:

```
notepad main.tf
```

Click:

Yes

when Notepad asks to create file.

Step 6 — Paste Terraform Code

Paste this properly:

```
terraform {  
  required_providers {  
    local = {  
      source = "hashicorp/local"  
      version = "~> 2.0"  
    }  
  }  
}
```

```
provider "local" {}
```

```
resource "local_file" "sample" {  
  filename = "sample.txt"  
  content = "Hello terraform"  
}
```

Save file.

What This Code Means

A. required_providers

`required_providers`

Tells Terraform which provider to download.

Here:

`hashicorp/local`

means:

Use local system resources.

B. provider "local"

`provider "local" {}`

Activates local provider.

C. resource "local_file"

`resource "local_file"`

Creates a file on your computer.

D. filename

`filename = "sample.txt"`

File name to create.

E. content

`content = "Hello terraform"`

Text written inside file.

Step 7 — Check File Exists

Run:

dir

You should see:

main.tf

Step 8 — Initialize Terraform

Run:

terraform init

What happens:

- Terraform downloads local provider plugin
- Creates .terraform folder

This is mandatory first step.

Step 9 — Check Plan

Run:

terraform plan

Terraform shows:

- What resource will be created
- Which file will be added

No actual changes yet.

Step 10 — Apply Configuration

Run:

terraform apply

Terraform asks:

Do you want to perform these actions?

Type:

yes

Press Enter.

Now:

Terraform creates:

sample.txt

Step 11 — Verify Output

Run:

dir

Now you will see:

sample.txt

Open it:

notepad sample.txt

Output:

Hello terraform

Step 12 — Destroy Resource

Run:

terraform destroy

Type:

yes

Terraform deletes:

sample.txt

MOST IMPORTANT CONCEPT

Terraform workflow:

Write Code

→ terraform init

→ terraform plan

→ terraform apply

→ Resource Created

→ terraform destroy

That is the real learning objective.

1. Install terraform
2. Add path:
windows + R → sysdm.cpl
3. Go to:
Advanced → ☒ Env Variables
select path
click edit
click New
Add → c:\Terraform
Then ok
4. CMD →
 1. terraform -version
 2. create project folder:
mkdir terraform-demo
cd terraform-demo.
 3. Create configuration file:
notepad main.tf → notepad mai code paste.
4. dir → to verify file exists)
 5. terraform init
 6. terraform plan
 7. terraform apply
yes
 8. explorer → to see file in folder. file explorer
 9. dir -force → check whats currently in folder.
 10. terraform destroy.

```

terraform {
  required_providers {
    local = {
      source = "hashicorp/local"
      version = "~> 2.0"
    }
  }
  provider "local" { }
  resource "local-file" "sample" {
    filename = "sample.txt"
    content = "Hello terraform"
  }
}

```

1. save time
2. reduce mistakes
3. Avoids rework.
4. improve reputation.